

Symmetry in a frozen galaxy encoder

Rotating a galaxy changes nothing physical about it. We re-encoded **512 real galaxies at 16 position angles** and mirrored, through frozen AION-1-Large, to ask what its image embedding does about orientation. Nothing in the AION-1 paper mentions rotation, flip or augmentation so there was no prior in either direction.

Every figure below is produced by a script in the repository and carries a provenance manifest recording its recipe, decisions and input hashes. In the live version each number is clickable and opens that record.

FINDINGS

1	The encoder is largely rotation-invariant, and nobody trained it to be Emergent, not designed. Measured directly from the rotation orbits.	80.9% POWER INVARIANT
2	What does vary is dominated by 180° symmetry m=2 is the signature of a two-fold symmetric shape a projected disc ellipse or a bar. m=4 (90°) is second at 17.9%.	37.7% OF VARYING POWER
3	That signal is real, not a resampling artefact 90° is the pixel grid's own symmetry, so m=4 could have been interpolation. Exact 90° turns are index permutations that resample nothing; compared against three resampled sub-orbits at identical geometry, they vary the same. Resampling accounts for 0.45%.	0.996 EXACT · RESAMPLED
4	Chirality is not encoded A mirrored galaxy lands on its own rotation orbit indistinguishable from a point known to be on it (0.974). Reflection is absorbed by rotation, so the model reads spiral arms (AUC 0.786) but not their winding direction.	0.973 MIRROR · NULL
5	Orientation costs ~2.4 of the manifold's ~14 dimensions Not the 4.32 a naive comparison gives: the full cloud is measured on 8,058 points and the quotient on 512, and TwoNN drifts with sample size. 1.94 of the apparent drop was estimator bias.	2.37 ± 0.60 DIMS
6	Orientation-free properties do <i>not</i> stay still Turning the same galaxy moves every readout by 0.20±0.42 of the probe's own error 0.46 dex peak-to-peak in stellar mass, 1.40 Gyr in age. The exception is bulge dominance at 0.0024: a radial concentration measure, and genuinely rotation-blind.	0.387 MASS WOBBLE · ERROR

HOW

512 anchor galaxies re-encoded at 16 angles (8,192 encodes) plus mirrored, through the shipped recipe which reproduces the distributed embeddings to cosine 0.999998. Orbits are decomposed into angular harmonics. **Real cutouts are 160 px and are rotated before the central 96 px crop**, so every rotated frame is fully defined; ASTRID cutouts are already 96 px with no margin, so only exact transforms apply there. Exact 90° multiples and mirrors go through `np.rot90` and index reversal pixel values unchanged, nothing resampled which is what makes findings 3 and 4 free of interpolation confounds. Finding 3 cost **zero GPU time**: the interpolation-free sub-orbit was already inside the 16-angle grid.

Finding 4's null matters as much as its number. "Close to the orbit" needs a reference measured the same way, so the angles are split into interleaved halves: the held-out half are points *known* to be on the orbit. Re-encoding a galaxy unflipped reproduces its stored vector to 1.2e-06 of the grid spacing, five orders of magnitude below the mirror distance so this is a real measurement, not numerical noise.

SECOND TRACK “ AN ENERGY-BASED MODEL

A parallel line of work asks the complementary question: not how the representation *moves* under transformation, but what it *holds*, and how tightly. An energy-based model is fitted over the AION-1 embedding and eight catalogue parameters, with an energy of the form $E = \sum a_i E_i(\text{emb}, x_i)$ “ one independent network per property, combined by learned coefficients that act as Lagrange multipliers on the components.

This differs from a linear probe in three ways that matter. It is **generative rather than discriminative**: embedding and parameters are modelled jointly, so properties compete for the same representation instead of each being regressed alone. Nested sampling over the energy returns a **posterior per parameter**, so “how well is this pinned down” is answered by a width rather than a single R^2 . And the learned a_i are themselves an interpretability readout “ a directly comparable ranking of how strongly the embedding is associated with each property “ while the per-parameter input projections show *which* embedding directions carry it.

Parameters covered: redshift, `logMstar`, `logMform`, `avgSfr`, `tageMw`, `zMw` and their errors. The model trains and samples; quantitative results were not finalised in time and are therefore not quoted here. Code is in `AION_explore.ipynb`.

LIMITS

- **Finding 6 is a conservative bound.** Those probes hold the penalty fixed rather than tuning it, so their residuals are larger than a tuned probe's “ which makes the ratios *smaller*. A better probe would look worse, not better.
- **16 angles resolves only to m=8** (Nyquist). m=4 already carries 17.9%, so power above m=8 could alias down. A 32-angle run was attempted; the 6 GB laptop GPU throttled to 270 MHz at 88 °C and was abandoned.
- **512 galaxies, not 10,000**, for everything needing re-encoding “ a GPU *time* limit, not memory: 6 GB caps the batch, throughput caps the count. That sample size is precisely what forced the correction in finding 5. The choice was justified against Ansuini et al. (TwoNN tracks ground truth at finite n below $ID\% \sim 20$) “ and our own measurement falsified it: the same cloud reads 14.62 at n=8,058 and 12.68 at n=512.
- **Chirality was tested geometrically, not against labels.** The orbit galaxies are from the anchor sample, whose morphology columns are 1.9–3.2% covered; the labelled `gzAim` sample is a different set of galaxies and the two are never merged.
- **Separately:** real vs simulated galaxies are perfectly separable (AUC 1.000) at every control stage “ but ASTRID has no PSF and no noise, so that is an upper bound on the physics gap, not a measurement of it.

TEAM & CODE

claude maxing “ Jack O'Brien (jackob@illinois.edu) • Murman Grugenidze (mgurgeni@andrew.cmu.edu)

Code, README and materialized outputs: `github.com/shturman99/KAAI_hackathon`